BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MIRPURKHAS

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A Multiple Choice Questions (MCQ's)

> Note: Q.No:01 Choose the correct Answer:

Qi. Which of the following is an alcohol?

(b) CH2CH=CH2
(b) CH3CH2CH3
(c) CH3-C=CH
(d) CH2=CH-C=CH

Qii. The prefi 'hept' stands for the ..... Carbon atoms.

(a) 2
(b) 5
(c) 7
(d) 9

Qiii. The functional grpup -COOH is used for:

(b) Alkynes
(b) alcohols
(c) Pehnols
(d) Carbohylic acids

Qiv. The carbohydrate s which contain aldehyde group are called:

(b) Sacchrides
(b) ketoses
(c) pentoses
(d) aldoses

Qv. Lactose is:

(b) Grape sugar
(b) honey sugar
(c) milk sugar
(d) cane sugar

Qvi. The unit of Kc for reaction $N_2 + O_2 = 2NO$

(a) mol dm^{-2}
(b) $mol^{1/2}$ dm^3
(c) mol^{-1} dm^3
(d) no unit

Qvii. The system is stable / equilibrium when:

(b) Qc = Kc
(b) Qc > K
(c) Qc < K
(d) None of these

Qviii. The value of Kc increases when:

(c) [Product] is less
(b) [Produce] is more
(d) [Reactant] is more
(d) Reactant = product]

Qix. NH_3 can be a base according to:

(a) Arrhenius theory
(b) Bronsted lowery theory
(c) Lewis
(d) both b and c

Qx. Which of the following is helpful for removal of permanent hardness:

(a) Na_2CO_3
(b) $Ca(OH)_2$
(c) $CaCO_3$
(d) Na_2SO_4

Qxi. The gas is mobile phase in:

(a) Liquid chromatography
(b) Solid chromatography
(b) Gas chromatography
(d) None of these

Qxii. The system is stable / equilibrium when:

(c) Qc = Kc
(b) Qc > K
(c) Qc < K
(d) None of these

Qxiii. The value of Kc increases when:

(e) [Product] is less
(b) [Produce] is more
(f) [Reactant] is more
(d) Reactant = product]

Qxiv. NH_3 can be a base according to:

(a) Arrhenius theory
(b) Bronsted lowery theory
(c) Lewis
(d) both b and c

Qxv. When the magnitude of Kc is small, indicates.

(a) Reaction mixture contain most of the reactant
(b) Reaction mixture contain most of the product
(c) Reaction mixture contain almost equal amount of reactant and product
(d) Reaction goes to completion

SECTION-B (Short Answers)

> Note: Answer any NINE of the following questions.

Q2. State law of mass action?

Q3. When writing an equation, how is a reversible reaction distinguished from irreversible reaction?

Q4. What is homologous series?

Q5. Define the functional group?

Q6. Distinguish between for salable and water soluble vitamins.

Q7. Elaborate the ionization equation of water.

Q8. Define buffers.

Q9. Define nomenclature and describe he I.U.P.A.C nomenclature rules for alkynes.

Q10. What is vitamin D? Give the sources and importance?

Q11. What are primary and secondary air pollutants.

Q12. Describe composition of water.

SECTION-C (Descriptive Answers)

> Note: Answer any THREE of the following questions.

Q13. Describe dynamic equilibrium with two examples?

Q14. Discuss the detail how different solutions in aqueous system exhibit increase in acidity and increase in basicity on the basis of H+ and OH?

Q15. Describe the uses of organic compound.

Q16. What are the carbohydrates? Explain sources and uses of carbohydrates.

Q17. Describe that how different air pollutants effects environment and human health?